COURSE PLAN FOR
CSET 207 COMPUTER NETWORKS

Faculty Names	: Mohd Abuzar Sayeed Amit Kumar Ankur Sharma Anuj Rai Arunabha Tarafdar Ashish Kumar Chanchal Maurya Debjani Ghosh Dipika Jain Gulshan Srivastava Hemant Kumar Saini Limbasiya Trupil G Neeraj Jain Nitin Shukla Nitin Tyagi Pawan Kumar Pranshu Saxena Prateek Gupta Pratibha Rajnish Kumar Chaturvedi Shivangi Nigam SimarPreet Singh Vidyapati jha Vimal Bagel Yogesh Pal Ravneet Kaur
Course Type	: Core
Semester and Year	: Summer Semester 2021-22
L-T-P	: 3-0-2
Credits	: 4
Course Level	: UG

School of Computer Science Engineering and Technology

BENNETT UNIVERSITY
THE TIMES GROUP

Greater Noida, Uttar Pradesh

COURSE CONTEXT

SCHOOL	**SCSET**	**VERSION NO. OF CURRICULUM/SYLLABUS THAT THIS COURSE IS A PART OF**	**V1**
DEPARTMENT		**DATE THIS COURSE WILL BE EFFECTIVE FROM**	**January, 2026**
DEGREE	**B.Tech.**	**VERSION NUMBER OF THIS COURSE**	**2**

COURSE BRIEF

COURSE TITLE	**Computer Networks**	**PRE-REQUISITES**	**NA**
COURSE CODE	**CSET207**	**TOTAL CREDITS**	**4**
COURSE TYPE	**Core**	**L-T-P FORMAT**	**3-0-2**

COURSE SUMMARY

This Course covers introduction to computer networks concepts, OSI and TCP/IP, Local area networks, Reliable data delivery, Routing and forwarding, Network applications and security in computer networks.

COURSE-SPECIFIC LEARNING OUTCOMES (CO)

By the end of this program, students should have the following knowledge, skills and values:

CO1: Examine the functionality of the different layers within network architecture.

CO2: Illustrate TCP/IP model suite protocols.

CO3: Design the networks for organization and select the appropriate networking architecture and technologies, subnetting and routing mechanism.

Detailed Syllabus

Module 1 (Contact Hours: 9)

Why Computer Networks: Applications of Networks, Transmission Media, Connecting Devices, Local Area Networks: LAN topologies: Bus topology, Ring topology, Star topologies, Mesh topology, Hybrid topology, OSI reference model, TCP/IP Protocol suite, Physical Layer: Services, Line coding scheme, Modulation, Multiplexing, Switching methods, Ethernet, Bluetooth, Wi-Fi, Wi- Fi Direct, WPA/WPA2/WPA3, Data Link layer: Services, Framing, Switches.

Module 2 (Contact hours: 8)

Reliable Data Delivery: Error detection, Error Correction, Flow control: Stop and wait, Go Back-N, Flow control: S-R Protocol, Error control (Retransmission techniques, timers), Medium Access sub layer - Channel Allocations, LAN protocols /ALOHA protocols, CSMA, CSMA/CD, Network Layer Protocols: Services (IP, ICMP), IP addressing, sub netting, Super netting (CIDR), IPV4, IPV6.

Module 3 (Contact hours: 9)

Routing and Forwarding, Static and dynamic routing, Unicast and Multicast Routing, Distance- Vector Routing, Link-State Routing, Shortest path computation-Dijkstra's algorithm, Address mapping-ARP, RARP, BOOTP, DHCP, Transport Layer: Services, UDP and TCP segment formats, connection establishment and termination, Expert Lecture from Industry, Congestion control, Congestion control: Open Loop and closed loop, Quality of service, Flow characteristics, Techniques to improve QoS.

Module 4 (Contact hours: 7)

Session Layer: Services, Protocols, Presentation layer: Services, Protocols, Application layer: Services, DNS, SIP, RTP, Telnet/SSH, HTTP, HTTPS, Remote login, Electronic mail, SMTP, FTP Commands and Replies, WWW, SNMP, Addressing Schemes, Uniform Resource Identifiers.

Module 5 (Contact hours: 9)

Principles of Cryptography, Symmetric key, Public key, Authentication protocols, Digital signatures, Firewalls, Security in different layers: Secure E-mail SSL, IP security, Advanced Topics in CN: Dark Net, CASS: Content-Aware Search System, Service-centric networking, Software-defined networking, Cloud Systems: Services, Data centre, 4G and 5G Networks, Body area sensor Networks, Satellite networks, SWARM networks.

STUDIO WORK / LABORATORY EXPERIMENTS:

Study of different types of networks cables and practically implement the cross-wired cable and straight through cable using clamping tool. Configure a network topology, connect different networks, static routing and dynamic routing, virtual LAN, RIP and OSPF using packet tracer. Also, Wireshark will be used for network troubleshooting, analysis, software, and communications protocol development.

TEXTBOOKS/LEARNING RESOURCES:

a) B. A. Forouzan, *Data communication and Networking* (5th ed.), McGraw Hill, 2021. ISBN 1260597822.

b) Andrew S. Tanenbaum and David J. Wetherall, *Computer Networks* (6th ed.), Pearson, 2021. ISBN 9780137523214.

REFERENCE BOOKS/LEARNING RESOURCES:

a) A. Trollope, *Data Communication & Networking* (1st ed.), PEARSON INDIA, 2020. ISBN 9781485832535.

EVALUATION POLICY

Sr. No	Component	Marks
1	Continuous Lab Evaluation	20
2	Class Participation Quiz	10
3	Certification	5
4	Assignment	5
5	Mid Term Exam	20
6	End Term Examination	40
	Total	**100**

CERTIFICATIONS

1. **The Bits and Bytes of Computer Networking.**
2. **CompTIA Network+ (N10-009) Crash Course Specialization.**
3. **CompTIA Network+ Certification (N10-009): The Total Guide Specialization.**
   ***All courses available on Coursera**